GGG-S-113
June 10, 1963
SUPERSEDING
(See 6.2)

FEDERAL SPECIFICATION

SCRAPER, BEARING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers scrapers for use in scraping bearings on both flat and curved surfaces.

1.1.1 *Federal specification coverage.* Federal specifications do not include all types and classes of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and classes.* Scrapers shall be of the following types and classes as specified (see 6.1):

Type I.—Triangle (three square).

Type II.—Half round (spoon).

Type III.—Flat.

Class 1.—End cutting, carbide tip.
Class 2.—End cutting, steel.

Type IV.—Double end, steel.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130—Identification Marking of U. S. Military Property.

FSC 5110

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Illustrations.** The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of scrapers which are otherwise in accordance with the requirements of this specification.

3.2 **Material.** All scraper blades, except type III, class 1, shall be forged of either a high-grade carbon or alloy steel. Type III, class 1 scrapers shall be as hereinafter specified.

3.2.1 *Handles.* Handles shall be of straight-grained hardwood, free from harmful defects. Moisture content shall not exceed 12 percent. Handles shall be of commercial design and shall be entirely suitable for the purpose intended. Each handle shall be fitted with a metal ferrule of sufficient strength and of the proper design to prevent splitting of the wood when the scraper tang is forced in the handle. Handles shall be securely attached to the tang of the scraper. Unless otherwise specified (see 6.1), handles shall be furnished on all scrapers, except type IV.

3.3 **Hardness.** Scraper blades, except type III, class 1, shall be hardened to not less than 58 nor more than 64 on the Rockwell "C" scale. Type III, class 1 blades shall be as hereinafter specified.

3.4 **Identification marking.**

3.4.1 *Civil agencies.* Scrapers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.4.2 *Military agencies.* Scrapers shall be marked in accordance with MIL-STD-130.

3.5 **Finish.**

3.5.1 *Blades.* Steel scraper blades shall be sharpened and ready for use. The edges shall be free of nicks and rough grind marks. The remainder of the blade shall have either an oil, ground, or enameled finish with no rough or sharp corners.

3.5.2 *Handles.* Handles shall be smoothly finished and shall have a protective coating of clear or colored varnish or lacquer.

3.6 **Type I, triangle (three square).** Type I scrapers shall be triangular shaped. The blade shall be hollow ground on all three sides to form the cutting edges. The blade, at the option of the manufacturer, may be relieved not more than one-half of the width of the blade. The scraper shall be straight from the tang, terminating in either a sweeping curve or tapering on all sides at least one-fourth of the blade length to form a point. Type I scrapers shall conform to table I and shall be similar to either option of figure 1.

Table I - Type I, triangle (three square).

Blade length ±1/2	Dimension across flats ±1/32	Handle length ±1/2	Handle diameter ±1/4
Inches	Inch	Inches	Inches
3	3/8	4	1
4	3/8	4	1
4	1/2	4-1/4	1-1/4
6	1/2	4-1/4	1-1/4
8	1/2	4-1/4	1-1/4
8	5/8	4-1/4	1-1/4

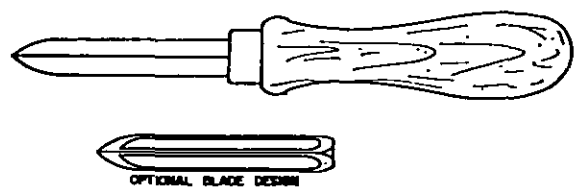


FIGURE 1.—Type I, triangle (three square).

3.7 **Type II, half round (spoon).** Type II scrapers shall consist of a spoon-shaped blade, a shank, and handle. The blade may be either hollow ground or relieved with

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

not less than 1/16 to 1/8 inch flat between the edge of the blade and the relief. The shank may be round or rectangular with the corners relieved of all burrs. Type II scrapers shall conform to table II and shall be similar to either option of figure 2.

Table II - Type II, half round (spoon).

Blade length ±1/8	Blade width at widest point		Shank length ±1	Handle length ±1/2	Handle diameter ±1/4
Inches	Inch		Inches	Inches	Inches
1-1/2	7/16	±1/32	2-3/4	4	1-1/8
2	7/16		3	4	1-1/8
2-1/2	9/16		3-3/8	4	1-1/4
3	5/8	±1/8	4	4-1/4	1-1/4
3-1/2	5/8		3	4-3/4	1-3/8
4	11/16		4	4-3/4	1-3/8
4-1/2	13/16		3	4-3/4	1-3/8

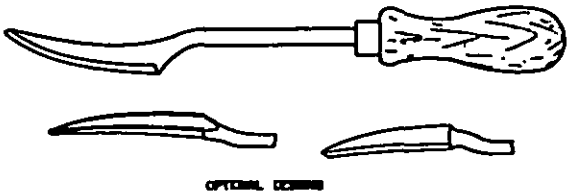


FIGURE 2.—Type II, half round (spoon).

3.8 Type III, flat.

3.8.1 *Class 1, end cutting, carbide tip.* Class 1 scrapers shall have a carbide tip silver soldered or brazed to a rolled steel shank. The steel shank shall have the corners rounded with not less than 1/16-inch radius. The carbide tip shall be in accordance with the carbide industry chart of grade recommendations for recommendation C-2. Both cutting edges of the tip shall be sharpened. The steel shank shall be hardened to not less than 28 nor more than 45 on the Rockwell "C" scale. Class 1 scrapers shall conform to table III and be similar to figure 3.

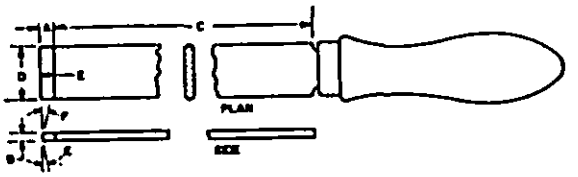


FIGURE 3.—Type III, class 1, flat, end cutting, carbide tip.

3.8.2 *Class 2, flat, end cutting, steel.* Class 2 scrapers shall be flat and shall have not less than 1/64-inch radius on the corners. Class 2 scrapers shall conform to table IV and be similar to figure 4.



FIGURE 4.—Type III, class 2, flat, end cutting, steel.

Table III - Type III, class 1, flat, end cutting, carbide tipped.

Tip length nominal A	Thickness ±0.010 B	Shank length nominal C	Width ±1/32 D	End radius nominal E	Angle (maximum permissible) F	Handle length ±1/2	Handle diameter ±1/8
Inch	Inch	Inches	Inch	Inches	Degrees	Inches	Inches
1/4	1/8	9-1/4	1	20	3	4-3/4	1-9/16

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:37Z
Check the source to verify that this is the current version before use.

Table IV - Type III, class 2, flat, end cutting, steel.

Blade length ±1/2	Blade thickness ±0.010	Blade width ±1/32	Handle length ±1/2	Handle diameter ±1/8
Inches	Inch	Inch	Inches	Inches
5	1/16	3/8	4-1/2	1-1/4
8	1/8	1	4-3/4	1-9/16

3.9 **Type IV, double end, steel.** Type IV scrapers shall be forged from either 1/2-inch diameter, round, or 1/2-inch across the flats, steel. Both ends shall taper approximately three inches to form the scraping edges. One end shall be ground square to the horizontal axis. The opposite end shall be convex. Type IV scrapers shall conform to table V and be similar to figure 5.

Table V - Type IV, flat, double end, steel.

Width of square end ±1/16	Width of convex end ±1/16	Radius of convex surface ±1/2
Inch	Inches	Inches
7/8	1-1/8	2



FIGURE 5.—Type IV, flat, double end, steel.

3.10 **Workmanship.** Scrapers shall be free of pits, seams, laps, heat checks, scale, excessive roughness, or any other defect that will impair its usefulness.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Sampling.**

4.2.1 *Inspection lot.* All scrapers of the same type or class presented at one time shall be considered a lot for purposes of acceptance inspection.

4.2.2 *Sampling for visual and dimensional examination.* Sample items shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the examination of 4.3. The Acceptable Quality Level (AQL) shall be 1.5 percent.

4.2.3 *Sampling for hardness tests.* Sample items shall be selected in accordance with inspection level I of MIL-STD-105. For the test of 4.4.1 the AQL shall be 4.0 percent defective.

4.3 **Visual and dimensional examination.** Each of the sample items selected in accordance with 4.2.2 shall be examined by the inspector to determine conformance with the visual and dimensional requirements of this specification.

4.4 **Test procedures.**

4.4.1 *Hardness.* The Rockwell hardness tests shall be conducted in accordance with method 243 of Fed. Std. No. 151. Care shall be exercised to assure that any decarburized surfaces are removed prior to the test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:37Z
Check the source to verify that this is the current version before use.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the scrapers shall be examined to determine compliance with the requirements of section 5 of this specification. Examination of the preservation and packaging requirements not covered by referenced specifications shall be performed in accordance with the quality assurance provisions of MIL-P-116. Examination of packing and marking requirements not covered by referenced specifications shall be in accordance with MIL-STD-105 using an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.

5.1.1 *Level A*. Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C*. Level C cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A*. Level A packing shall be in accordance with level A of MIL-H-15424.

5.2.2 *Level B*. Level B packing shall be in accordance with level B of MIL-H-15424.

5.2.3 *Level C*. Scrapers, cleaned, preserved, and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies*. In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies*. Interior packages and shipping containers shall be marked in accordance with MIL-H-15424.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, and style required (1.2).

(*c*) Blade length and width required (applicable tables).

(*d*) Whether scrapers should be furnished without handles (3.2.1).

(*e*) Level of preservation, packaging, and packing required (5.1 and 5.2).

(*f*) Marking, if different (5.3).

6.2 Supersession data. This specification replaces MIL-S-001618B(SHIPS), dated 7 October 1959 and MIL-S-16186A, dated 3 March 1952.

6.3 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:
Tools, hand, not otherwise indexed.
Truckload minimum weight (w) 30.6 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implications or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—GL

Navy—Sh

Air Force—MOAMA

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.